

AYMANE AIT DADS

AI Intern | Generative AI · Retrieval-Augmented Generation · Agentic AI Workflows

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and executing an end-to-end LoRA SFT pipeline on a 30B-parameter model - directly demonstrating enterprise-grade generative AI development capability. Reduced manual network analysis time by 60% at Orange Maroc by deploying production ML pipelines on 100K+ records, translating model outputs into actionable stakeholder recommendations with measurable operational impact. Brings hands-on experience in RAG architecture, agentic pipelines, LLM orchestration via Claude API and LangChain, backend REST API integration, and cloud data infrastructure using Supabase and PostgreSQL - aligning precisely with Hendrickson's AI intern requirements. Aims to accelerate Hendrickson's enterprise AI adoption by applying proven fine-tuning, retrieval-augmented generation, and agentic workflow design skills to real production systems.

CORE COMPETENCIES

Generative AI Development | Retrieval-Augmented Generation | Agentic AI Workflows | Backend API Development | Cloud Deployment | Enterprise Data Integration | LLM Orchestration | Prompt Engineering

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Developed an **automated end-to-end ML pipeline** combining Random Forest classification and K-Means clustering on 100K+ fiber-optic network records, reducing manual analysis time by approximately 60%.
- Built a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with outputs adopted directly by the network optimization team for production maintenance decisions.
- Engineered features from raw upload speed, latency, and jitter signals across **thousands of network nodes**, delivering structured model evaluation reports consumed by non-technical stakeholders.
- Delivered stakeholder presentations translating complex **model outputs into maintenance recommendations**, bridging data science results and enterprise operational decision-making using Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)

Kaggle Competition · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) targeting all attention projections, MLP layers, and lm_head.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce 605 and 64 verified chain-of-thought rows respectively.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection - EURECOM + NTIRE 2026 @ CVPR

EURECOM ImSecu Course · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class private Kaggle leaderboard (0.791 AUC)** and submitted to the NTIRE 2026 @ CVPR CodaBench international benchmark by fine-tuning a CLIP ViT-L/14 (428M parameters) with a custom MLP head and dual learning-rate strategy.
- Trained on **250K images spanning 25+ generator types** with FP16 mixed precision, 10-view TTA, and a 3-model ensemble weighted by validation AUC, discovering that 1-epoch training (0.793 AUC) outperforms 4-epoch training for out-of-distribution generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Cloud Computing, Web Applications, Image Security, Statistics

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

AI & LLM Engineering: Generative AI | RAG | LLM Orchestration | LangChain | Agentic Pipelines | Claude API | Prompt Engineering
ML Frameworks & Backend: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | LoRA | SFTTrainer | vLLM | Scikit-learn
Cloud, Data & APIs: Supabase | PostgreSQL | REST APIs | Docker | Python | SQL | Power BI | Pandas | Git